



UNIVERSITI MALAYSIA TERENGGANU
FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

CSE3433

Software Architecture

Cloud-Native Learning Management System (LMS)
(Cloud-Native / Layered Architecture)

Prepared by:

Group No.5

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Prepared for:

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COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONOUR
SEMESTER II 2025/2026

WEEKLY LOG ENTRIES

PHASE 1: Project Proposal

WEEK 5

Undertaken Tasks & Weekly Activities	● Organised the official project meeting for Group 5 to establish our development lifecycle roadmap and evaluated our cloud deployment options to ensure system scalability. ● Finalised technical stack selection: Railway for hosting and automatically scaling our application services, AWS (Amazon Web Services) to supply heavy duty cloud database and storage infrastructure and Apache NetBeans as our shared local development IDE. ● Compiled and formatted all components of Deliverable 1 into Project Proposal document.
Individual Contributions (as Project Manager):	● Drafted structural foundations of the report, establishing problem description regarding monolithic LMS bottlenecks and laying down the system objectives. ● Authored team member roles and responsibilities section and mapping individual strengths to specific architectural system boundaries.
Challenges Encountered	Figuring out how to map our multi platform cloud setup cleanly into traditional layered and microservices grading criteria required for the course.
Action Plans	Made sure we clearly documented in the proposal how using Railway for the application layer and AWS for the data layer enforces strict isolation of concerns and satisfies cloud-native architecture paradigms.

Table 1 : Weekly Progress Log (Week 5)

PHASE 2: Architecture Design

WEEK 6

Undertaken Tasks & Weekly Activities	• Coordinated a team meeting to break down Deliverable 2 requirements, focusing on identifying core LMS modules and defining non functional metrics like scalability and system availability. • Managed initial workspace coordination, planning out how our application services built in Apache Netbeans will securely link to our cloud backend hosting and infrastructure.
Individual Contributions (as Project Manager):	Outlined the system overview section of the report, clearly maps the data boundaries and traffic flow between our live application instance on Railway and our AWS cloud data resources.
Challenges Encountered	Team confusion about database synchronization and how to connect multiple local Apache NetBeans IDE environments to a single, live AWS cloud database instance.
Action Plans	Established a unified environmental variable protocol to manage secure cloud connection strings across the team without causing configuration conflicts.

Table 2 : Weekly Progress Log (Week 6)

WEEK 7

Undertaken Tasks & Weekly Activities	Checked through the final versions of our system context, component, interaction and deployment diagrams before compiling them with the final text into our Deliverable 2 report for submission.
Individual Contributions (as Project Manager):	Wrote the section explaining our architecture style and did a final check on all four diagrams to make sure our split deployment across Railway and AWS was clearly represented before we submitted the document.
Challenges	It was quite tricky trying to show a modern multi provider

Encountered	cloud infrastructure using old-school UML deployment nodes since our application layers run on Railway and our data resources runs as a cloud service instead of local servers.
Action Plans	I decided we should represent our setup by drawing Railway as our application deployment runtime platform node and then mapping out AWS as a separate, interconnected external infrastructure block broken down into our specific database and storage components so it makes sense visually.

Table 3 : Weekly Progress Log (Week 7)

PHASE 3: Project Observation 1 & Prototype Progress

WEEK 8

Undertaken Tasks & Weekly Activities	Held a team meeting to prepare for Project Observation 1 and started laying out our frontend components in Apache NetBeans to connect them with cloud servers
Individual Contributions (as Project Manager):	Organized project checklist for presentation and sure application files were ready for an initial mock setup.
Challenges Encountered	It was hard trying to plan out how the real-time notifications would sync up smoothly between our applications services hosted on Railway and our AWS cloud database.
Action Plans	Use optimize API endpoints or native database features within AWS PostgreSQL to ensure the system updates data automatically whenever changes occur.

Table 4 : Weekly Progress Log (Week 8)

Tool Used	ChatGPT
Purpose	Brainstorming how real-time components work across distributed cloud hosting environments.
Prompt Use	"How do I explain real-time data synchronization features between a web service hosted on Railway and a cloud database running on AWS?"
Output Summary	Explained how structured API polling, persistent WebSocket connections and managed database listener triggers handle live updates efficiently without heavy performance overhead.
How it used	Used this information to help explain our real time notification synchronization logic during our team planning session.

Table 5 : GenAI Usage Record (Week 8)

WEEK 9

Undertaken Tasks & Weekly Activities	Presented our project architecture and initial deployment plans during the Project Observation 1 session with our lecturer.
Individual Contributions (as Project Manager):	Led presentation by walking through our choice of utilizing Railway and AWS and took down the feedback we received to improve our system flow.
Challenges Encountered	The feedback highlighted that we needed to make sure our user authentication records were completely isolated and secured from normal, public facing course management data.
Action Plans	I worked with Aida to ensure we configure separate, protected database schemas within our AWS PostgreSQL database instead of throwing all credentials into a standard public data table.

Table 6 : Weekly Progress Log (Week 9)

Tool Used	Gemini
Purpose	Formatting presentation points based on feedback.
Prompt Use	"How to structure database user isolation strategies between application services and AWS RDS PostgreSQL after an architectural review feedback."
Output Summary	Recommended using strict PostgreSQL role privileges, separate database schemas and keeping application auth tokens secure.
How it used	Used this layout approach to plan the database security adjustments with Aida

Table 7: GenAI Usage Record (Week 9)

WEEK 10

Undertaken Tasks & Weekly Activities	Monitored the core prototype progress as Aida finished setting up the main database tables on AWS and Syaheera prepared the login screens to deploy on Railway.
Individual Contributions (as Project Manager):	I mapped out the integration test cases to ensure that testing the login system wouldn't mess up our existing user records in the live AWS database instance.
Challenges Encountered	We noticed a mismatch where some data types created in NetBeans didn't line up properly with the PostgreSQL data types inside our cloud database.
Action Plans	I had the team map out a quick data-type matching guide to keep our data formats consistent across our local code and the cloud host environments

Table 8 : Weekly Progress Log (Week 10)

Tool Used	Gemini
Purpose	Looking up data type conversions for database mapping.
Prompt Use	"What is the standard matching data type in PostgreSQL on AWS RDS for a basic string array coming from an application built in NetBeans?"
Output Summary	Recommended using the standard text [] array type or JSONB for flexible object schemas.
How it used	Passed this tip to Aida so she could adjust her table column properties inside our AWS PostgreSQL database.

Table 9: GenAI Usage Record (Week 10)

WEEK 11

Undertaken Tasks & Weekly Activities	We ran our final integration tests on the prototype and successfully demonstrated our initial working system for Deliverable 3.
Individual Contributions (as Project Manager):	Gathered in GitHub repository links and took screenshots of our working UI, Railway deployment streams and AWS logs to package everything into a submission file.
Challenges Encountered	Right before submission, the live deployment on Railway threw an error because some critical cloud database environment variables were missing in the online project console
Action Plans	Manually logged into Railway dashboard settings and added the missing AWS connection strings to fix the live application environment deployment.

Table 10 : Weekly Progress Log (Week 11)

Tool Used	Gemini
Purpose	Creating clear technical documentation for a project submission.
Prompt Use	"Give me a clean layout format to present multi-provider cloud deployment screenshots (Railway and AWS) and repository links for an academic system prototype report."
Output Summary	Provided a neat template with a clear description block, system environment notes, and source links.
How it used	Used this to format our Deliverable 3 submission document.

Table 11: GenAI Usage Record (Week 11)

PHASE 4: Project Observation

WEEK 12

Undertaken Tasks & Weekly Activities	We presented our integrated system for Project Observation 2, demonstrating how our live application layers deployed on Railway communicate with our AWS cloud infrastructure resources.
Individual Contributions (as Project Manager):	Managed presentation flow for the session and structured our team checklist to handle final feedback regarding external service hosting.
Challenges Encountered	We noticed that initial database requests from our services on Railway to our live AWS RDS PostgreSQL instance were experiencing slight network latency bottlenecks
Action Plans	Worked with Aida to look into connection pooling and guided the team to optimize server environment variables handle long-lived connections between Railway and AWS.

Table 12 : Weekly Progress Log (Week 12)

Tool Used	ChatGPT
Purpose	Troubleshooting cross-cloud latency issues.
Prompt Use	"How do I minimize database connection latency between an application hosted on Railway and a PostgreSQL database running on AWS RDS?"
Output Summary	Suggested enabling connection pooling, optimizing indexing on foreign keys, and ensuring both environments are configured in matching geographic cloud regions if possible
How it used	Used this advice to help Aida tweak our connection strings in our Apache NetBeans configuration setup.

Table 13: GenAI Usage Record (Week 12)

WEEK 13

Undertaken Tasks & Weekly Activities	Collected and organized all our individual technical writes, diagram updates and environment setups to compile and submit the final project report
Individual Contributions (as Project Manager):	Formatted the entire final report layout, making sure technology stack section clearly and accurately documented our deployment boundaries across Railway and AWS.
Challenges Encountered	Combining deep technical explanations about Railway deployment with AWS security policies from sections written by my teammates.
Action Plans	Editing session rewriting the transitions to smooth out the tone, ensuring the entire document reads seamlessly as a single, high-quality academic paper.

Table 14 : Weekly Progress Log (Week 13)

Tool Used	Gemini
Purpose	Organizing a clean academic report layout.
Prompt Use	"How do I neatly structure an implementation and deployment summary section for a university report when using multiple cloud providers like Railway and AWS?"
Output Summary	Recommended using a clear tabular breakdown or distinct sub headers dividing the compute/hosting layer from the data persistence layer.
How it used	Used formatting suggestion to organize our report Technology Stack and Deployment sections cleanly.

Table 15: GenAI Usage Record (Week 13)